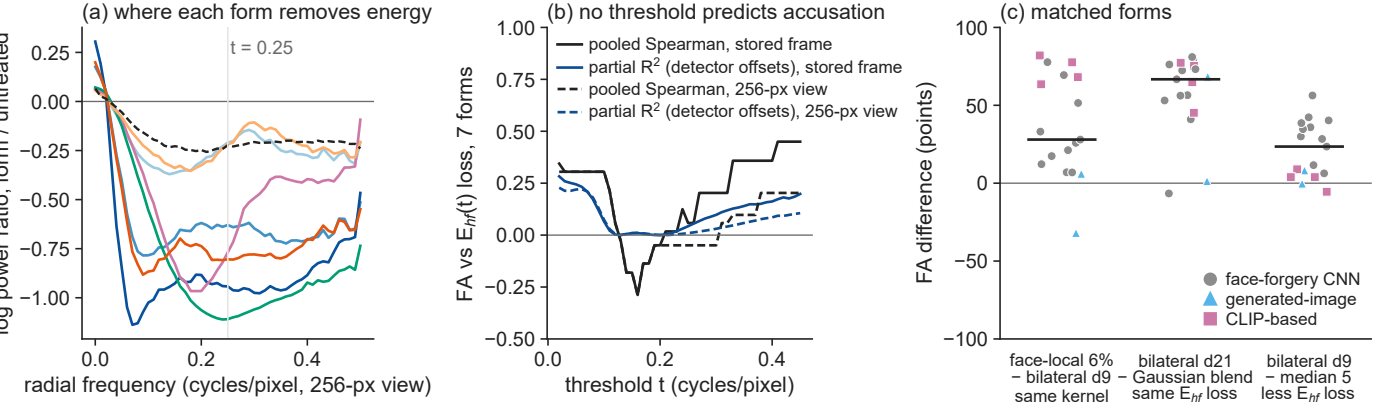
(a) where each form removes energy
t = 0.25
log power ratio, form / untreated
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.0
0.2
0.4
radial frequency (cycles/pixel, 256-px view)
(b) no threshold predicts accusation
pooled Spearman, stored frame
partial R² (detector offsets), stored frame
pooled Spearman, 256-px view
partial R² (detector offsets), 256-px view
FA vs E_hf(t) loss, 7 forms
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
0.0
0.2
0.4
threshold t (cycles/pixel)
(c) matched forms
FA difference (points)
100
50
0
−50
−100
face-forgery CNN
generated-image
CLIP-based
face-local 6% − bilateral d9 same kernel
bilateral d21 − Gaussian blend same E_hf loss
bilateral d9 − median 5 less E_hf loss
bilateral d9
bilateral d15
bilateral d21
median 5
Gaussian blend
face-local 6%
face-local 12%
service S-90